

For each limit of the reference interval, the MU is calculated and added as a “guard band” represented with horizontal red stripes. Consequently, the differentiation zone between males and females is significantly reduced as illustrated by the dashed rectangle.

This study was undertaken to detect the use of testosterone as a doping agent for athletes. Therefore, the reference intervals for three groups of pathological individuals are added to the graphics as rectangles with crossed stripes, namely 5- α -reductase deficiency type 2 (5ARD2), androgen insensitivity syndrome (AIS) and the polycystic ovary syndrome (PCOS).

It is obvious that it is impossible to distinguish between healthy people who may have used testosterone and a certain number of pathologic individuals. The testosterone test presents a real case that illustrates classic difficulties of decision-making, either to determine the pathology of an individual, or, in this example, to control an illicit substance intake. Many reference values are one-sided and should be considered as a lower or upper limit not to be exceeded. The reference values are not the same in different countries making the decision even more difficult to make using a given biological marker.

In the area of food safety, the development of reference values, such as Maximum Residue Limits (*MRL*) established by the European Commission [3] and illustrated in Section 9.3 is somewhat more complex because it requires the combination of three sources of data:

- Toxicological reference values (TRV). These are toxicological thresholds established by international or national authorities such as the World Health Organization (WHO) that reflect a dose–response relationship or a point calculated from such a relationship. In the dietary risk analysis, TRVs are specifically estimated for oral ingestion in water and food, and most often for low level exposure over extended periods, such as lifetime. Most classical naming of the TRV is acceptable daily intake (ADI) or tolerable daily intake (TDI) when considering intentionally added substances, such as additives.
- Food composition data for nutrients and contaminants. There are many databases available online. However, there are two problems related to the collection of this type of data. On the one hand, the very incomplete state of knowledge on the composition of foods, since a food as *simple* as wine contains nearly 250 chemical constituents, including alcohol and water. On the other hand, the notion of “average” composition is very misleading since data significantly vary and are not distributed according to a normal law. This last remark is even more exact for contaminants and pollutants.
- Food consumption data. It is probably the most difficult information to obtain because the collection cost is high and the representativity poorly guaranteed. There are multiple strategies for investigating individual or family diet over varying periods of time. They all present biases, and it is by crossing different surveys that it is possible to reach some accuracy.

The three types of data are combined to define exposure levels. As the data are not normally distributed, such studies raise complex statistical issues which have been

previously addressed by one of us [4]. Commonly applied methods allow to identify the most probable sources of risk, i.e. main foods which are the key-sources of risk. It is then up to the regulator to decide which specification limit should apply for which analyte in which food.

It is a rather complex process based on much data, which explains why sanitary limits can often be modified. Another body in charge of this mission is the Codex Alimentarius, which also publishes internationally accepted *MRLs* [5]. Local regulation may also put forward other thresholds. But if the country is a member of the WTO, these local thresholds must be justified and not appear as technical barriers to trade (TBT).

Like the European Union, Codex also defines performance requirements for the methods that will be used to enforce the established specification limits. This task is specifically assigned to the Codex Committee on Methods of Analysis and Sampling or CCMAS. For example, for a method to be acceptable it must fulfill following requirements:

If the <i>MRL</i> is,	then the LOQ must be
$\geq 0.1 \text{ mg/kg}$	$\leq \frac{MRL}{5}$
$< 0.1 \text{ mg/kg}$	$\leq \frac{2 \times MRL}{5}$

The Codex Alimentarius approach is missing a guideline on the procedure to be applied by laboratory in demonstrating that the proposed method meets required criteria and properly proves its ability to perform an inspection.

8.1.3 Role of the Analytical Report

The analytical report is another essential element in the relationship between the laboratory and its client facing a decision. It is a contractual document and therefore has a formal aspect complementary to the technical work of the analyst. The content is described in documents relating to accreditation or quality control.

For example, in the context of the accreditation of medical laboratories, ISO 15189 standard introduces the concepts of pre-analytical, analytical, and post-analytical procedures. The reception of an analysis request sent by a prescriber or the sampling form is the pre-analytical part. The analysis report or report of the result is the post-analytical part. As addressed in the ISO 17025 and 15189 standards, elaborating the analytical report appears to be a formal administrative task. In fact, it involves the civil liability of the laboratory.

To ascertain this responsibility, a set of statements must appear in the report, like the regulatory name of the analyte, the unit used, etc. This is perfectly detailed in the regulatory texts, standards, or guides provided by the accreditation bodies. Inspection bodies are very vigilant about this documentary and/or regulatory aspect. The requirements may vary according to the accreditation program. In general, it is required that the expression of results be unequivocal.

For example, the method of analysis used must be mentioned, and even reagents, whenever they may affect the expression of the result, as well as when required by regulations. Many statements seem odd, such as: “For quantitative results, where appropriate, the analytical performance of the method may be stated.” Does this mean that for nonquantitative methods, such as identification assays or diagnostic tests, the mention of sensitivity and specificity is superfluous? The analysis report is printed on a laboratory-headed paper, including the regulatory references, and is signed.

It can only be communicated after the procedure called *analytical validation*, which is different from the *method validation* described in Chapter 5. Analytical validation is carried out under the responsibility of the head of the analytical service. It consists in checking if the conditions of execution of the analysis conform with preestablished specifications. It can only be carried out after checking the indicators of instrument qualification and the results of the internal quality control. A complementary *scientific* validation can also be carried out under the responsibility of a accredited specialist in most of the laboratories. This involves ensuring the consistency of the results of all the analyses performed.

Up to now, the indication of the MU in a significant part of the analytical report is not required. While ISO 15189 does not mention it, ISO 17025 (clause 7.8.3) states that:

“where appropriate, the measurement uncertainty expressed in the same unit as the measurand or in a term related to the measurand (e.g. as a percentage), where it is significant for the validity or application of the test results, where required by customer instructions, or where the uncertainty affects conformity with the limits of a specification.”

Similar remarks can be found in the documents written by various accreditation authorities.

8.2 Sample Conformity Assessment

8.2.1 Define the Decision Rule

Conformity assessment refers to the “activity to determine whether specified requirements relating to a product, process, system, person or body are filled.” The issue of conformity assessment is formally addressed in the standard ISO 17025:2017 by means of a clearly stated decision rule defined as a (clause 3.7) “rule describing how measurement uncertainty is accounted when stating conformity with a specific requirement.”

The same standard describes the context of the application of this rule (clause 7.1.3) “when the customer requests a statement of conformity ... the decision rule must be communicated to and agreed with the customer, unless inherent in the requested specification of standard.” Moreover, (clause 7.8.6.1) “when a statement

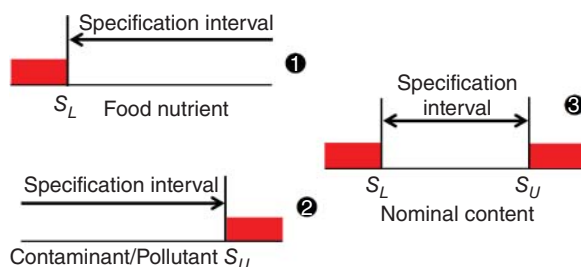


Figure 8.2 Three ways to assess sample conformity to a unilateral or bilateral specification interval.

of conformity ... is provided, the laboratory must document the decision rule it employs, taking into account the level of risk associated with the decision rule, and apply the decision rule" but no further explanation is given on how to practically apply the decision rule.

One of the most frequent examples of a decision is the verification of manufactured lot conformity. In other words, the demonstration that the concentration of a target analyte measured in a sample (that may contain several pooled sampling units) complies with the value of a limit of specification, i.e. a threshold. According to the situation, it can be a lower specification limit, denoted S_L , an upper specification limit S_U , or, more rarely, a specification interval $[S_L, S_U]$; the first two define a unilateral interval, the last a bilateral interval.

Figure 8.2 illustrates these three types of intervals. For example, for the control of pollutants or contaminants, the concentration must remain below an upper limit S_U , which classically the legislator has defined, such as the European regulation 470/2009 "laying down Community procedures for the establishment of residue limits of pharmacologically active substances in foodstuffs of animal origin" [6].

Compliance is therefore declared in relation to the specification interval. Usually, the decision rule is more complicated than simply looking at one reading. It is often established in relation to the sampling plan, and several measurements may be used. The sampling plan was previously established to compute what is called the efficiency of the plan, i.e. the probability of detecting a nonconforming lot.

This efficiency may vary depending on the number of accepted defects. For instance, it can be a zero-defect plan or where a certain number of results nonconforming to the specification interval are acceptable. It is not the purpose here to discuss further the theory of sampling plans, as many books and several series of guidelines describing decision rules for measurement or attribute-based control plans for different decisional contexts, such as industrial production of clinical biology, exist.

In many laboratories, a common procedure is to proceed in steps. First, an initial measurement is made. When the latter is not within the acceptability interval, so that the sample should be declared nonconforming, it is accepted to obtain a confirmatory replicate. If the first measurement is invalidated, a third one will be done.

From there, the decision rule can be fundamentally modified (and not always reported). This iterative procedure raises various problems, including statistical, that are rarely addressed. For example, when the decision rule is based on a single result, it is a direct violation of the rule. Or, when the result is given as an

average of replicates, the calculation of the associated MU must involve an adjusted intermediate precision standard deviation because these replicates are generally obtained in intermediate precision conditions. A discussion on the consequences of measurement replication on uncertainty is presented in Section 8.4.3.

8.2.2 Guard Band Concept

In 2012, the BIPM published a document known as JCGM-106 on the role of MU in conformity assessment [7]. Most of the concepts developed here have been taken up in the general ISO guide 98-4:2012 or the EURACHEM guide specifically dedicated to using MU for compliance assessment using chemical measurements [8, 9].

An important concept introduced in this JCGM document is the “guard band” that corresponds to the expanded uncertainty estimated for the concentration at the limit of specification. It is added or subtracted to the limit of specification depending on whether it is a lower or upper specification limit.

Figure 8.3 illustrates this concept and associated vocabulary in the case where an upper limit of specification must be checked, for example, for pollution control. To run a control, the main issue is calculating the guard band. In the example of Figure 8.3, it is equal to the expanded uncertainty of the specification limit, which should cover 95% of future measurements around this limit. If the uncertainty function of the method is available, this calculation is easy. This approach seems simple but is not without problems in its application to the analytical sciences because the uncertainty varies rapidly with the concentration. Even for a small range of concentrations, MU can differ significantly. This property of the MU can have important consequences for the guard band calculation, as illustrated by the following simulated data.

Let's assume two different compliance objectives. In the case illustrated on the left, Figure 8.4, the objective is an upper limit that should not be exceeded. The decision rule is simple: if the analysis reveals that the concentration is too high, the batch is rejected. On the right side of Figure 8.4, it is a minimum content that must be reached, and the decision rule is reversed. In both cases, the limit of the specification is 5.00 in arbitrary units. The interpretation of this figure is as follows:

- The specification limit appears as a thick vertical line.
- The limit of the guard band is marked by a dashed vertical line.

Figure 8.3 The guard band concept introduced by the JCGM.

